**CIS 21031 – PRACTICAL FOR PLATFORM TECHNOLOGIES**
**DEPARTMENT OF ICT**
**FACULTY OF TECHNOLOGY**
**SOUTH EASTERN UNIVERSITY OF SRILANKA**

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**Labsheet:** 04

**Date:** 01.12.2023

**Title:** Programming Shell Scripts- II

**Aim:** Getting practice to write a shell script with arithmetic operations and user input commands.

**Tasks:**

- Working with arithmetic operations in bash.
- Working with user inputs in bash.

**Exercise 01: Create a shell script Exercise1.sh**

1. Write a shell script which could do the following arithmetic operations using ***'let'*** expression.

   *Basic format:*

   ***let <arithmetic expression>***

   a. Initialize that x=5+6.
   b. Print the value of x.
   c. Increment the value of x by 1 and print its value
   d. Next, print the value of x.
   e. Initialize that a=14, b=8 and print the value of y=a-b.
   f. Decrement the value of y by 2 and print its value.
   g. Find the product of 4*5 by initializing it to a variable called z, and then print its value.

***2.*** Write a shell script which could do the following arithmetic operations using ***double parentheses*** and do the following arithmetic operation.

   *Basic format:*

   ***$((  expression ))***

   a) Addition
   b) Subtraction

c) Multiplication
d) Division
e) Modular

3. Write a shell script which could do the following arithmetic operations using ***'expr'*** expression.

   *Basic format:*

   ***$( expr item1 math-operator item2 )***

   a) Define three variables a, b and c and assign values 2, 5, 9 respectively.
   b) Print the three variables.
   c) Next print "c % b equals to <<value of c MOD b >>".
   d) Compute 'c minus b' and save it to another variable 'P'. Print 'P'
   e) Print a * c equals to <<value of a * c >>".
   f) Finally, print "c/b equals to << value of c/b >>".

4. Write the script *to return the length of following variables*.

   *Basic format:*

   ***${#variable}***

   a) Declare a variable called a and make its value as "Hello World".
   b) Print the length of a.
   c) Declare a variable called b and make its value as '1234' and print the length of b.

**Exercise 02: User Inputs in bash**

a) Create the shell script 'Exercise2.sh', ask user for their name and print the message "It's nice to meet you <<name>>.
b) Ask user for login details such as username and password and print "Thankyou <<username>> you have now entered your login details."

c) Create a shell script to display the following information.

- Print “What is your first name?” and get user input.
- Print “What is your degree program?” and get input from user.
- Request for the Registration Number.
- Prompt the user to enter a User name.
- Next, request the user to enter a password (The password should not be visible to any users).
- Then print “Hello <<first name>> .Your Registration number is <<Reg. Number>>. You are reading <<degree>> degree program and your user name is <<User Name>>”.

**<u>Practice Question</u>**

1. Create the shell script arithmetic1.sh and do the following arithmetic operations using ***‘expr’.***

   i. Define two variables **a** and **b** and assign values 8 and 10, respectively.
   ii. Print the two variables.
   iii. Then print sum of the two variables.
   iv. Next print **a** MOD **b**.
   v. Compute **a** minus **b** and save it to another variable ‘**z’**. Print ‘**z’**.
   vi. Print **a** * **b**.
   vii. Finally, print **a** / **b**.

2. Create the shell script arithmetic2.sh and do the following arithmetic operations using ***‘let’.***

   i. Initialize that **k**=2+4.
   ii. Print the value of **k**.
   iii. Use quotes and again initialize **k**=3+6.
   iv. Next, print the value of **k**.
   v. Increment the value of **k** by 1 and print its value.

vi. Find the product of 5*5 by initializing it to a variable called **a**, and then print its value.

3. Create the shell script arithmetic3.sh and do the following arithmetic operations using ***'double parentheses'.***

   i. Initialize that 7+3 is **r**, and print the value of **r**.
   ii. Check whether are you able to get the same output by spacing it out.
   iii. Assign a variable called **n** and perform the summation of **r**+5, and the output for **n**.
   iv. Increment the value of **n** by 1 and display the output.
   v. Increment the value of **n** by 3 and display the output.

4. Create the shell script 'convert.sh' and write a program for feet and inches conversion.
   i. Print a number from user input command.
   ii. Convert this number into inchers and feet.
   (*Hint1: number feet to inches: number*12*)
   (*Hint2: number inches to feet: number/12*)
   *(Hint3: refer arithmetic operation for floating point and **use bc command**)*
   iii. Discuss the bc command.

***Submit your lab record on or before 5th of December 2023.***